

# Notes: Ausubel (AER, 1991)

## The Failure of Competition in the Credit Card Market

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# 1 Big picture

- **Question.** Why did the U.S. bank credit card market fail to look competitive in the 1980s, even though it appeared to have thousands of issuers, low entry barriers, national solicitation, and little direct price regulation?
- **Central paradox.** The market looks like a textbook case where competition should discipline prices: roughly 4,000 issuing banks, about 75 million consumers, no obvious scarce input, and no clear evidence of explicit collusion. Yet credit card interest rates remain close to 18% while banks' cost of funds moves substantially.
- **Main price result.** Credit card rates are extremely sticky. In regressions of credit card rates on lagged rates and lagged cost of funds, the coefficient on the cost of funds is only about 0.04–0.05, far from the near-one response suggested by a continuous spot-market competitive benchmark.
- **Main profit result.** Major issuers earn very high returns. For large credit card banks, adjusted pretax return on equity is often in the 70%–120% range during 1983–1988, compared with an ordinary pretax banking return near 20%.
- **Ex ante confirmation.** Credit card portfolios sell between banks at large premia over outstanding balances. The average reported resale premium is about 20%, inconsistent with a view that observed profits are merely accidental ex post realizations.
- **Consumer-behavior mechanism.** The paper argues that many consumers choose cards without fully internalizing the high probability that they will later borrow and pay finance charges. These consumers are not very responsive to interest-rate differences ex ante, even though they later become profitable borrowers.
- **Adverse-selection logic.** A bank that lowers its rate attracts a disproportionate share of borrowers who plan to revolve balances and are likely to be worse credit risks. This discourages price competition on the finance charge and pushes competition toward nonprice dimensions.
- **Policy message.** The paper does not simply say “regulate rates.” It argues that the laissez-faire defense based on perfect competition is empirically weak, while conventional usury ceilings also create distortions by excluding risky borrowers. The welfare problem is therefore real but not mechanically solved by price caps.
- **Broader takeaway.** Many-firm market structure is not enough for competition. Demand-side inattention, search or switching frictions, and adverse selection can make a seemingly competitive consumer-credit market behave as though prices are sticky and profits persistent.

## 2 Market, data, and measurement (key objects)

### 2.1 The market: why competition seemed plausible

The paper studies U.S. bank-issued MasterCard and Visa credit cards. The institutional setting initially looks competitive for several reasons:

- there are thousands of issuing banks;

- the top ten issuers account for only about 30% of active accounts and about 43% of outstanding balances in 1987;
- many issuers solicit customers nationally;
- after the *Marquette* decision and subsequent state-law changes, major issuers can effectively avoid binding state usury ceilings by locating credit card operations in states with permissive rules;
- there is no obvious input scarcity, major sunk cost, or explicit cartel arrangement.

*Interpretation.* The paper’s argument is powerful because it starts from a market that should have been friendly to the competitive model. If even this market produces sticky prices and high returns, the issue is not merely a mechanical lack of firms.

## 2.2 Issuing banks are the relevant firms

Visa and MasterCard are not treated as the main competitive units. The issuing banks own the accounts and choose key consumer-facing contract terms, including:

- the annual percentage rate,
- annual fees,
- grace periods,
- credit limits,
- late fees and other account terms.

*Interpretation.* The relevant product-market question is therefore not whether there are two card networks. It is whether issuing banks compete away profits in the pricing of card credit.

## 2.3 The central price measure

The main price is the credit card interest rate charged on revolving balances. The aggregate series comes from the Federal Reserve’s quarterly survey of banks. The paper also collects bank-level price data through the author’s Bank Credit Card Survey (BCCS).

The cost-of-funds measure is:

$$CoF_t = y_t^{1yr\ Tbill} + 0.75\%, \quad (1)$$

where the 0.75 percentage-point spread approximates the yield spread on credit-card-backed securities over Treasury securities of comparable maturity.

*Interpretation.* This cost measure is meant to be conservative. If credit card receivables are short-duration assets and securitized receivables trade close to Treasury yields plus a modest spread, then large credit card spreads cannot easily be explained by funding cost alone.

## 2.4 The Bank Credit Card Survey

The BCCS is a hand-collected survey of large bank credit card issuers. The first survey generated pricing and loss data from 21 banks, including an interest-rate series for 17 issuers and a loan-loss

series for 10 issuers. Follow-up surveys obtained direct profit calculations from seven large issuers.

*Interpretation.* These data are unusual because they separate credit card operations from the rest of banking. The paper is not only inferring profitability from aggregate bank accounts; it uses direct survey evidence, Call Report evidence, prospectus information, and portfolio-sale prices.

## 2.5 Profitability variables

The paper measures profitability using pretax return on assets and pretax return on equity. Since the capital requirement during the period is roughly 6% of assets, the paper maps adjusted return on assets into adjusted return on equity by:

$$ROE_{it}^{adj} = \frac{ROA_{it}^{adj}}{0.06}. \quad (2)$$

An ordinary pretax banking return is taken to be about:

$$ROA^{ord} \approx 1.20\%, \quad ROE^{ord} \approx \frac{1.20\%}{6\%} = 20\%. \quad (3)$$

*Interpretation.* This normalization is central. A credit card ROA of 5% or 6% may not sound enormous, but if only about 6% equity capital supports the asset, it maps into very large returns on equity.

## 2.6 Adjusted ROA

The adjusted ROA measure replaces each bank's reported interest expense with a standardized funding cost. A simplified version is:

$$ROA_{it}^{adj} = \frac{Profit_{it} + InterestExpense_{it} - 0.94 \cdot CoF_t \cdot Assets_{it}}{Assets_{it}}. \quad (4)$$

The factor 0.94 appears because roughly 6% of assets are funded by equity, so only the nonequity portion is assigned a market funding cost.

*Interpretation.* This adjustment prevents low-cost deposit funding from mechanically making the credit card business look profitable. If a bank's branches generate cheap deposits, that value should not be attributed entirely to credit cards.

## 2.7 Portfolio resale premia

The paper also studies prices paid when one bank buys another bank's credit card accounts. Let outstanding balances equal  $X$ . If a portfolio sells for  $(1 + \Phi)X$ , then  $\Phi$  is the premium.

- In a competitive zero-profit long-run equilibrium, the portfolio should sell near par.
- If accounts are expected to generate persistent excess profits, the buyer should be willing to pay a premium.

*Interpretation.* This is especially useful because it is an ex ante market valuation. Portfolio buyers are paying today for expected future profits.

## 2.8 Consumer-behavior data

The BCCS asks banks what fraction of customers avoid finance charges and what average balances active accounts carry. The data show that only about 21%–32% of customers avoid finance charges in the years reported, while average outstanding balances rise from about \$523 in 1979 to about \$1,038 in 1987.

*Interpretation.* Most active customers are not pure convenience users. A large majority pay interest at high rates, yet many consumers appear not to shop as though they expect to borrow.

## 3 Models and empirical specifications (all equations + interpretation)

### 3.1 Competitive benchmark

A simple competitive benchmark says the finance charge should move closely with marginal lending cost:

$$r_t = MC_t + markup_t, \quad (5)$$

where  $r_t$  is the credit card interest rate and  $MC_t$  is the marginal cost of credit, dominated by the cost of funds and expected losses.

In a spot-market competitive equilibrium with a stable risk environment, changes in the cost of funds should pass through substantially into card rates:

$$\frac{\partial r_t}{\partial CoF_t} \approx 1. \quad (6)$$

*Interpretation.* The paper does not require a zero spread. Credit cards are risky and costly to service. The problem is that spreads remain very large and rates barely move when funding costs change.

### 3.2 Sticky-rate regression

The basic price-adjustment regression is:

$$r_t = \alpha + \rho r_{t-1} + \beta CoF_{t-1} + \varepsilon_t. \quad (7)$$

A bank-level version adds bank dummy variables:

$$r_{it} = \alpha_i + \rho r_{i,t-1} + \beta CoF_{t-1} + \varepsilon_{it}. \quad (8)$$

The paper reports  $\beta \approx 0.042$  in the aggregate Federal Reserve series and  $\beta \approx 0.054$  in the bank-level BCCS series.

*Interpretation.* A one-percentage-point increase in the cost of funds moves card rates by only about 4–5 basis points in the short run. The high coefficient on lagged card rates implies substantial inertia. This is the formal version of the figure showing almost flat credit card rates while funding costs vary widely.

### 3.3 Nonprice competition and charge-offs

The industry explanation for high spreads was that credit losses rose. Ausubel's counterargument is that causality may run the other way. If rates stay high and banks compete by expanding credit to riskier customers, then charge-offs can rise because the high spread makes it profitable to lend to marginal borrowers.

A simple way to state the logic is:

$$d^* \approx r - c, \quad (9)$$

where  $d^*$  is the marginal default risk the bank is willing to accept,  $r$  is the interest rate, and  $c$  is the nondefault marginal cost of funds and servicing.

*Interpretation.* If  $r - c$  rises, banks may move down the credit-quality distribution. That makes loan losses partly endogenous to the high rate spread rather than an exogenous justification for it.

### 3.4 Profit accounting

For a credit card portfolio, the basic accounting identity is:

$$Profit = FinanceCharges + Fees + InterchangeRevenue - InterestExpense - OperatingExpense - NetChargeOffs. \quad (10)$$

Return on assets is:

$$ROA = \frac{Profit}{OutstandingBalances}. \quad (11)$$

Adjusted ROE is then:

$$ROE^{adj} = \frac{ROA^{adj}}{0.06}. \quad (12)$$

*Interpretation.* The credit card business earns large profits not because any single line item is mysterious. Finance charges are very large relative to funding costs and losses, so even after fees, operating costs, and charge-offs, the residual is enormous.

### 3.5 Ex ante portfolio valuation

The resale-premium calculation capitalizes expected excess returns. In the paper's notation, if an account lives for  $N$  years, outstanding balances grow at rate  $g$ , the discount rate is  $r$ , credit card ROA is  $ROA_{cc}$ , and ordinary banking ROA is  $ROA_{ord}$ , then the implied premium is:

$$\Phi = (ROA_{cc} - ROA_{ord}) \sum_{k=0}^{N-1} \left( \frac{1+g}{1+r} \right)^k. \quad (13)$$

*Interpretation.* A premium is the present value of expected excess profits on the acquired customer base. The higher the expected persistence or growth of balances, the larger the premium justified by a given excess ROA.

### 3.6 Competitive-rate calculation

Let  $s$  be the share of balances that actually accrue interest. If actual adjusted ROA exceeds ordinary ROA by  $ROA^{adj} - ROA^{ord}$ , then the interest-rate cut that would eliminate excess profits is approximately:

$$\Delta r \approx \frac{ROA^{adj} - ROA^{ord}}{s}. \quad (14)$$

For 1987, the paper uses  $ROA^{adj} \approx 5.72\%$ ,  $ROA^{ord} \approx 1.20\%$ , and  $s \approx 0.90$ , giving:

$$\Delta r \approx \frac{5.72\% - 1.20\%}{0.90} \approx 5.02 \text{ percentage points}. \quad (15)$$

Thus, with an actual rate around 18.67%, the implied competitive rate is about:

$$r^{comp} \approx 18.67\% - 5.02\% = 13.65\%. \quad (16)$$

*Interpretation.* This is not presented as a precise regulatory formula. It is a back-of-the-envelope way to express the magnitude of excess pricing: roughly five percentage points in the late 1980s.

### 3.7 Search and switching costs

Search and switching frictions can make prices sticky. Consumers may face costs of finding lower-rate cards, filling out applications, waiting for approval, losing remaining annual-fee value, or worrying about credit limits and credit records.

*Interpretation.* Search and switching costs can explain why some customers remain with high-rate issuers. But the paper argues that observed resale premia are so large that search/switch costs alone may be insufficient unless one assumes very high consumer frictions.

### 3.8 Adverse selection and consumer inattention

The paper's main behavioral mechanism divides consumers into three broad groups:

- **Unintentional revolvers:** consumers who do not plan to borrow but repeatedly end up paying finance charges. These are highly profitable and relatively insensitive to interest rates ex ante.
- **Intentional revolvers:** consumers who expect to borrow, are more rate-sensitive, and may be worse credit risks.
- **Convenience users:** consumers who never borrow and are mainly sensitive to annual fees, rewards, and grace periods.

If a bank cuts its interest rate, it attracts mainly the intentional revolvers rather than the profitable inattentive borrowers. The adverse-selection effect discourages rate cutting.

*Interpretation.* The key idea is that the most profitable customers are not the most price-sensitive. That breaks the usual competitive force that low rates should win profitable business.



### 3.9 Regulation model

Let  $P_n$  be the bank's estimate of consumer  $n$ 's default probability and let  $c$  be the marginal cost of funds and servicing exclusive of default. A personalized efficient rate would be:

$$r_n^* = P_n + c. \quad (17)$$

A conventional usury ceiling sets one cap  $\bar{r}$ . Then consumers with:

$$P_n > \bar{r} - c \quad (18)$$

may be denied credit, while lower-risk consumers may all pay rates close to the cap.

*Interpretation.* Rate caps can reduce rents, but they can also ration riskier borrowers out of the market and flatten risk-based pricing. The policy problem is therefore a tradeoff, not a simple command to impose ceilings.

## 4 Identification, credibility, and interpretation

### 4.1 Why the evidence is hard for perfect competition

The paper offers several mutually reinforcing facts:

- price stickiness: rates barely respond to cost-of-funds changes;
- high accounting profits: large issuers earn returns far above ordinary banking returns;
- robustness across sources: survey data, Call Reports, prospectuses, and functional-cost data point in the same direction;
- ex ante valuation: portfolio-sale premia suggest market participants expected profits to persist;
- consumer evidence: most active cardholders pay finance charges, even though many consumers claim or behave as if they do not expect to borrow.

*Interpretation.* Each piece alone could be challenged. Together, they make a coherent case that the competitive model is missing something important.

### 4.2 Why concentration is not the main story

The market is not highly concentrated in the usual sense. The top ten issuers hold less than half of outstanding balances, and many banks can legally solicit customers across states. Therefore, the paper does not rely on a standard high-Herfindahl monopoly story.

*Interpretation.* The puzzle is precisely that market structure looks competitive while market outcomes do not.

### 4.3 Why explicit collusion is not the main story

The paper reports no evidence of explicit collusion on price or quantity. Moreover, some large issuers cut rates in the mid-1980s without triggering a market-wide price war; other issuers maintained high rates without obvious loss of their customer bases.

*Interpretation.* This pattern is more consistent with customer frictions and adverse selection than with a simple cartel story.

### 4.4 Why loan losses do not fully explain high rates

Loan losses increase in the mid-1980s, but the magnitude is modest relative to the spread between credit card rates and funding costs. The paper also argues that charge-offs can be endogenous: high spreads induce banks to approve riskier accounts.

*Interpretation.* Rising losses are not ignored. They are incorporated into the discussion, but they do not eliminate the evidence of excess profitability.

### 4.5 Why ex post profitability is not enough, and how the paper responds

A natural objection is that 1983–1988 may have been an unusually favorable period. The paper responds in three ways:

- functional-cost evidence suggests smaller issuers also earned above-normal returns over a longer window;
- loan-loss data from earlier downturns do not show losses large enough to erase the profit story;
- portfolio-sale premia show that buyers expected future profits, not merely that sellers happened to have had good past outcomes.

*Interpretation.* The resale evidence is particularly important because it speaks to ex ante expectations.

### 4.6 Accounting concerns

The paper adjusts interest expense to avoid attributing cheap-deposit rents to credit cards. It also triangulates across several data sources: direct survey responses, FDIC Call Reports, SEC prospectuses, and observed portfolio sale prices.

*Interpretation.* No accounting exercise is perfect, but the same broad conclusion appears across independent sources.

### 4.7 What the paper does not fully prove

The paper makes a strong case that perfect competition fails as a descriptive model. It does not fully quantify the exact contribution of each mechanism. Search costs, switching costs, consumer inattention, adverse selection, rewards, annual fees, and nonprice competition may all matter.

*Interpretation.* The paper’s core empirical contribution is the documented anomaly. Its theoretical contribution is to show why consumer behavior and adverse selection can rationalize that anomaly.

## 5 Tables and figures

### 5.1 Table 1: market structure

**Read:**Table 1 lists the largest MasterCard and Visa issuers in 1987. The top ten issuers have about 35.6 million accounts, or 30% of accounts, and about \$40.7 billion in balances, or 43.4% of outstanding balances.

**Economic message.** The market is not dominated by a single firm or a tiny cartel. That makes sticky rates and high profits more surprising.

TABLE 1—TOP TEN ISSUERS OF MASTERCARD AND VISA CARDS, 1987

| Bank                     | Number of accounts | Percentage market share (by number of accounts) | Outstanding balances (\$ billion) | Percentage market share (by outstanding balances) |
|--------------------------|--------------------|-------------------------------------------------|-----------------------------------|---------------------------------------------------|
| Citibank                 | 10,000,000         | 8.4                                             | \$15.3B                           | 16.3                                              |
| Chase Manhattan Bank     | 5,000,000          | 4.2                                             | \$5.4B                            | 5.8                                               |
| Bank of America          | 4,800,000          | 4.0                                             | \$5.2B                            | 5.5                                               |
| First Chicago            | 4,500,000          | 3.8                                             | \$4.6B                            | 4.9                                               |
| Manufacturers Hanover    | 3,300,000          | 2.8                                             | \$2.0B                            | 2.1                                               |
| Wells Fargo Bank         | 1,800,000          | 1.5                                             | \$2.8B                            | 3.0                                               |
| Maryland Bank            | 1,800,000          | 1.5                                             | \$1.7B                            | 1.8                                               |
| Marine Midland Bank      | 1,700,000          | 1.4                                             | \$1.4B                            | 1.5                                               |
| Chemical Bank            | 1,500,000          | 1.3                                             | \$1.3B                            | 1.4                                               |
| Associates National Bank | 1,200,000          | 1.0                                             | \$1.0B                            | 1.1                                               |
| Top ten                  | 35,600,000         | 30.0                                            | \$40.7B                           | 43.4                                              |
| Second ten               | 11,500,000         | 9.7                                             | \$9.0B                            | 9.6                                               |
| Total                    | 118,900,000        | 100.0                                           | \$93.9B                           | 100.0                                             |

*Sources:* Individual banks’ numbers of accounts surveyed by *American Banker*, (March 1, 1988, pp. 1–2) and *Credit*

**Figure 1:** Table 1: Top issuers of MasterCard and Visa cards, 1987. Source: Ausubel (1991).

### 5.2 Figure 1: sticky credit card rates

**Read:**Figure 1 compares the credit card interest rate with the cost of funds from 1982 to 1989. The cost of funds falls and rises substantially, while the credit card rate remains nearly flat around 18%.

**Economic message.** This is the paper’s most direct visual evidence against rapid competitive pass-through. A competitive market can have a spread, but it should not ignore large changes in funding costs for years.

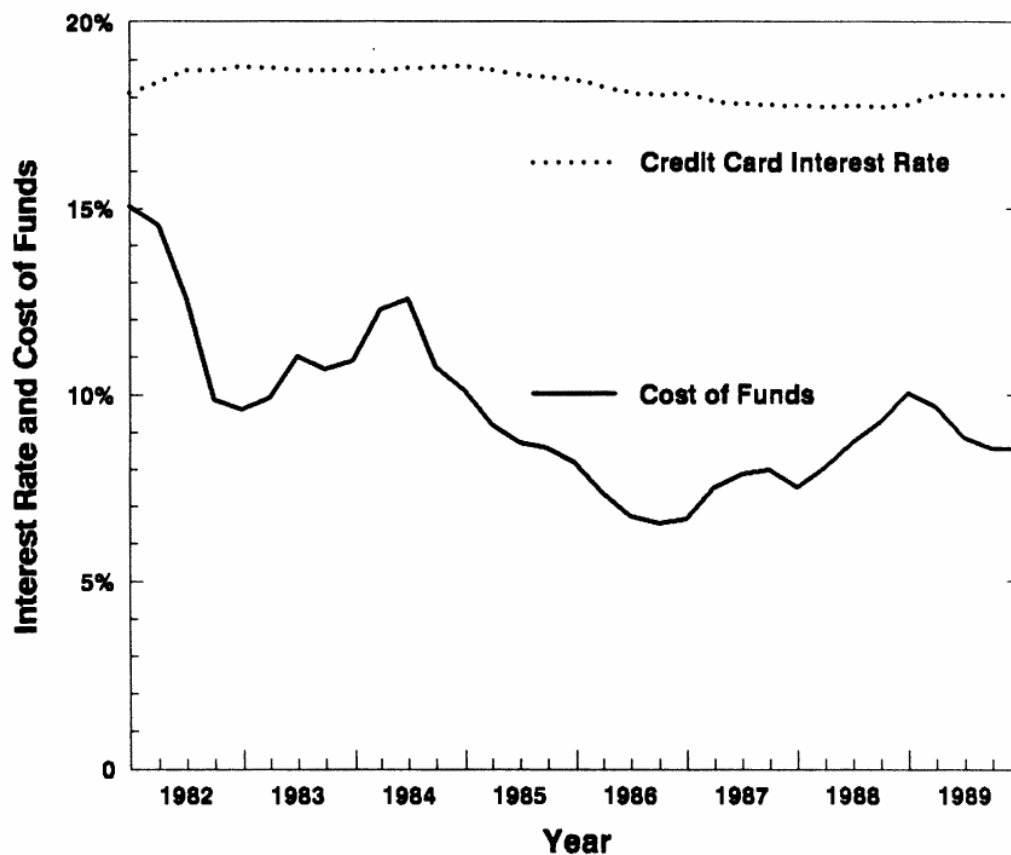


FIGURE 1. STICKY CREDIT CARD INTEREST RATES, 1982–1989

Figure 2: Figure 1: Sticky credit card interest rates, 1982–1989. Source: Ausubel (1991).

### 5.3 Table 3: regression evidence on stickiness

**Read:** Table 3 estimates the relationship between credit card rates, lagged card rates, and lagged cost of funds. The coefficient on cost of funds is 0.0422 in the Federal Reserve series and 0.0540 in the bank-level survey series.

**Economic message.** Even statistically significant cost pass-through is economically tiny. The lagged credit card rate dominates the regression, meaning the best predictor of today's credit card rate is yesterday's credit card rate, not today's funding environment.

TABLE 3—ORDINARY LEAST-SQUARES REGRESSION  
OF CREDIT CARD INTEREST RATE ON COST OF  
FUNDS AND LAGGED CREDIT CARD INTEREST RATE  
(QUARTERLY, 1982–1987)

| Variable                                      | Federal Reserve<br>Board<br>survey data | Bank<br>credit card<br>survey data |
|-----------------------------------------------|-----------------------------------------|------------------------------------|
| COST OF<br>FUNDS <sub>-1</sub>                | 0.0422<br>(0.00584)                     | 0.0540<br>(0.00896)                |
| CREDIT CARD<br>INTEREST<br>RATE <sub>-1</sub> | 0.895<br>(0.0444)                       | 0.685<br>(0.0326)                  |
| Constant                                      | 1.51<br>(0.807)                         |                                    |
| Bank-1 dummy                                  |                                         | 5.75<br>(0.659)                    |
| Bank-2 dummy                                  |                                         | 5.11<br>(0.594)                    |
| Bank-3 dummy                                  |                                         | 6.38<br>(0.719)                    |
| Bank-4 dummy                                  |                                         | 5.79<br>(0.657)                    |
| Bank-5 dummy                                  |                                         | 5.70<br>(0.651)                    |
| Bank-6 dummy                                  |                                         | 4.18<br>(0.500)                    |
| Bank-7 dummy                                  |                                         | 5.69<br>(0.645)                    |
| Bank-8 dummy                                  |                                         | 5.12<br>(0.595)                    |
| Bank-9 dummy                                  |                                         | 5.61<br>(0.644)                    |
| Bank-10 dummy                                 |                                         | 5.44<br>(0.634)                    |
| Bank-11 dummy                                 |                                         | 5.00<br>(0.589)                    |
| Bank-12 dummy                                 |                                         | 5.12<br>(0.595)                    |
| Bank-13 dummy                                 |                                         | 4.99<br>(0.586)                    |
| Bank-14 dummy                                 |                                         | 4.88<br>(0.577)                    |
| Bank-15 dummy                                 |                                         | 5.50<br>(0.636)                    |
| Bank-16 dummy                                 |                                         | 5.22<br>(0.593)                    |
| Bank-17 dummy                                 |                                         | 6.17<br>(0.707)                    |
| Number of<br>observations:                    | 24                                      | 408                                |
| $R^2$ :                                       | 0.96                                    | 0.937                              |
| Durbin $h$ :                                  | -0.69                                   | 0.10                               |

Notes: CREDIT CARD INTEREST RATE is the dependent variable in each regression. COST OF FUNDS is defined as the yield on one-year Treasury bills plus 0.75 percent. Observe that there is no cross-firm variation in this variable, so that year dummy variables cannot be included in the second regression equation.

Figure 3: Table 3: Interest-rate regression. Source: Ausubel (1991).

## 5.4 Table 4: charge-offs and spreads

**Read:** Table 4 shows charge-offs and interest-rate spreads for 1976–1987. Charge-offs rise in the mid-1980s, but spreads are also very high.

**Economic message.** Loan losses are part of the cost of credit, but they do not fully explain the high and sticky spreads. Moreover, when spreads are high, issuers may profitably extend credit to

riskier borrowers, making losses partly an outcome of pricing rather than a pure cause.

**TABLE 4—LOAN LOSSES ON CREDIT CARDS OF  
TEN SURVEY BANKS DURING 1976–1987  
(FEWER THAN NINE DURING 1976–1978 AND 1986–1987)  
AND THE SPREAD BETWEEN CREDIT CARD  
INTEREST RATES AND THE COST OF FUNDS**

| Year | Average charge-off<br>rate<br>(percentage) | Interest rate<br>spread<br>(percentage) |
|------|--------------------------------------------|-----------------------------------------|
| 1976 | 1.15                                       | 10.57                                   |
| 1977 | 0.99                                       | 10.36                                   |
| 1978 | 1.27                                       | 8.11                                    |
| 1979 | 1.44                                       | 5.78                                    |
| 1980 | 2.04                                       | 5.08                                    |
| 1981 | 1.48                                       | 2.94                                    |
| 1982 | 1.67                                       | 6.42                                    |
| 1983 | 1.32                                       | 8.79                                    |
| 1984 | 1.36                                       | 7.69                                    |
| 1985 | 1.94                                       | 9.82                                    |
| 1986 | 3.01                                       | 11.55                                   |
| 1987 | 2.60                                       | 10.43                                   |

*Source:* Author's bank credit card survey (Appendix A, Table A1, questions 7 and 1).

**Figure 4:** Table 4: Loan losses and credit card spreads. Source: Ausubel (1991).

## 5.5 Tables 6 and 7: direct and Call Report profitability

**Read:** Table 6 reports direct profit data from seven issuers. Average adjusted ROE equals 88.7% in 1984, 114.9% in 1985, 118.9% in 1986, 105.2% in 1987, and 83.6% in 1988. Table 7 reports Call Report evidence: average adjusted ROE is 74.0% in 1983 and remains well above ordinary banking returns through 1988.

**Economic message.** The core profit evidence is not a small-sample anecdote. Different data sources show annual returns multiple times ordinary banking returns.

TABLE 6—RETURN ON ASSETS AND RETURN ON EQUITY (PRETAX) BASED ON DIRECT REPORTS OF CREDIT CARD ISSUERS

| Bank                    | Rank  | Measure        | Percentage returns |       |       |       |       |
|-------------------------|-------|----------------|--------------------|-------|-------|-------|-------|
|                         |       |                | 1984               | 1985  | 1986  | 1987  | 1988  |
| A                       | 1–10  | ROA (reported) | 5.98               | 4.36  | 5.59  | 6.91  | N.A.  |
|                         |       | ROA (adjusted) | 6.09               | 6.06  | 6.64  | 6.53  | N.A.  |
|                         |       | ROE (adjusted) | 101.5              | 101.1 | 110.7 | 108.8 | N.A.  |
| B                       | 11–20 | ROA (reported) | 8.09               | 7.78  | 3.74  | 4.21  | N.A.  |
|                         |       | ROA (adjusted) | 8.00               | 7.22  | 7.15  | 5.04  | N.A.  |
|                         |       | ROE (adjusted) | 133.3              | 120.3 | 119.1 | 84.0  | N.A.  |
| C                       | 11–20 | ROA (reported) | 5.01               | 5.70  | 3.82  | 4.55  | 4.30  |
|                         |       | ROA (adjusted) | 5.00               | 6.21  | 6.49  | 5.33  | 4.23  |
|                         |       | ROE (adjusted) | 83.3               | 103.5 | 108.2 | 88.8  | 70.5  |
| D                       | 21–30 | ROA (reported) | 7.20               | 7.86  | 7.81  | 7.93  | 8.05  |
|                         |       | ROA (adjusted) | 3.28               | 6.39  | 7.33  | 6.39  | 5.84  |
|                         |       | ROE (adjusted) | 54.7               | 106.5 | 122.2 | 106.5 | 97.4  |
| E                       | 21–30 | ROA (reported) | 8.48               | 8.92  | 8.74  | 9.96  | 9.69  |
|                         |       | ROA (adjusted) | 7.48               | 9.20  | 10.02 | 8.09  | 7.31  |
|                         |       | ROE (adjusted) | 124.6              | 153.3 | 166.9 | 134.8 | 121.8 |
| F                       | 31–40 | ROA (reported) | 7.27               | 7.54  | 6.41  | 8.39  | 5.87  |
|                         |       | ROA (adjusted) | 5.75               | 6.62  | 5.91  | 6.98  | 4.95  |
|                         |       | ROE (adjusted) | 95.9               | 110.3 | 98.5  | 116.4 | 82.5  |
| G                       | 41–50 | ROA (reported) | 2.24               | 6.37  | 6.26  | 6.15  | 2.75  |
|                         |       | ROA (adjusted) | 1.67               | 6.54  | 6.41  | 5.81  | 2.74  |
|                         |       | ROE (adjusted) | 27.8               | 109.0 | 106.8 | 96.9  | 45.6  |
| Direct report averages: |       | ROA (reported) | 6.32               | 6.93  | 6.05  | 6.87  | 6.13  |
|                         |       | ROA (adjusted) | 5.32               | 6.89  | 7.14  | 6.31  | 5.01  |
|                         |       | ROE (adjusted) | 88.7               | 114.9 | 118.9 | 105.2 | 83.6  |

Source: Author's bank credit card survey (Appendix A, follow-up survey, Table A2, question 4).

Figure 5: Table 6: Direct survey evidence on credit card profitability. Source: Ausubel (1991).

TABLE 7—RETURN ON ASSETS AND RETURN ON EQUITY (PRETAX) BASED ON CALL REPORTS FILED WITH THE FDIC

| Bank                              | Rank            | Measure          | Percentage returns |       |       |       |      |       |
|-----------------------------------|-----------------|------------------|--------------------|-------|-------|-------|------|-------|
|                                   |                 |                  | 1983               | 1984  | 1985  | 1986  | 1987 | 1988  |
| Citibank (South Dakota), N.A.     | 1               | ROA (reported)   | 7.44               | 7.05  | 6.02  | 6.24  | 5.31 | 4.26  |
|                                   |                 | ROE (actual cap) | 81.4               | 80.3  | 75.5  | 75.1  | 77.7 | 48.6  |
|                                   |                 | ROA (adjusted)   | 5.40               | 5.24  | 5.56  | 6.62  | 5.61 | 3.92  |
|                                   |                 | ROE (adjusted)   | 90.1               | 87.4  | 92.7  | 110.3 | 93.5 | 65.4  |
| Chase Manhattan Bank (U.S.A.)     | 2               | ROA (reported)   | 4.73               | 5.63  | 5.29  | 6.03  | 4.50 | 3.84  |
|                                   |                 | ROE (actual cap) | 56.4               | 71.2  | 109.1 | 146.5 | 75.2 | 53.9  |
|                                   |                 | ROA (adjusted)   | 3.32               | 3.62  | 4.28  | 5.46  | 3.24 | 2.68  |
|                                   |                 | ROE (adjusted)   | 55.3               | 60.4  | 71.3  | 91.1  | 54.0 | 44.7  |
| Maryland Bank, N.A.               | 7               | ROA (reported)   | 8.14               | 7.35  | 7.09  | 6.63  | 4.80 | N.A.  |
|                                   |                 | ROE (actual cap) | 74.8               | 142.0 | 121.6 | 128.8 | 86.8 | N.A.  |
|                                   |                 | ROA (adjusted)   | 7.72               | 6.76  | 8.04  | 7.67  | 4.86 | N.A.  |
|                                   |                 | ROE (adjusted)   | 128.6              | 112.7 | 134.0 | 127.8 | 81.0 | N.A.  |
| Beneficial National Bank (U.S.A.) | 15 <sup>a</sup> | ROA (reported)   | 4.23               | 5.10  | 3.20  | 1.61  | N.A. | N.A.  |
|                                   |                 | ROE (actual cap) | 34.3               | 77.7  | 39.6  | 22.0  | N.A. | N.A.  |
|                                   |                 | ROA (adjusted)   | 1.31               | 4.01  | 4.27  | 3.70  | N.A. | N.A.  |
|                                   |                 | ROE (adjusted)   | 21.9               | 66.9  | 71.1  | 61.7  | N.A. | N.A.  |
| Lomas Bank U.S.A.                 | 18              | ROA (reported)   | N.A.               | N.A.  | N.A.  | 3.59  | 4.74 | 4.35  |
|                                   |                 | ROE (actual cap) | N.A.               | N.A.  | N.A.  | 60.6  | 80.7 | 39.9  |
|                                   |                 | ROA (adjusted)   | N.A.               | N.A.  | N.A.  | 4.47  | 5.20 | 4.71  |
|                                   |                 | ROE (adjusted)   | N.A.               | N.A.  | N.A.  | 74.5  | 86.6 | 78.5  |
| CoreStates Bank of Delaware       | 20              | ROA (reported)   | 2.86               | 4.21  | 5.09  | 6.46  | 5.03 | 4.10  |
|                                   |                 | ROE (actual cap) | 63.9               | 61.9  | 100.1 | 131.8 | 95.1 | 70.1  |
|                                   |                 | ROA (adjusted)   | 2.71               | 4.65  | 6.14  | 7.55  | 5.91 | 4.57  |
|                                   |                 | ROE (adjusted)   | 45.1               | 77.6  | 102.3 | 125.9 | 98.6 | 76.2  |
| First City Bank—Sioux Falls       | 21              | ROA (reported)   | N.A.               | N.A.  | 6.88  | 4.15  | 5.34 | 6.85  |
|                                   |                 | ROE (actual cap) | N.A.               | N.A.  | 74.0  | 41.6  | 43.4 | 48.5  |
|                                   |                 | ROA (adjusted)   | N.A.               | N.A.  | 5.82  | 4.56  | 5.59 | 6.58  |
|                                   |                 | ROE (adjusted)   | N.A.               | N.A.  | 97.1  | 75.9  | 93.2 | 109.6 |
| First Omni Bank, N.A.             | 33              | ROA (reported)   | 8.68               | 5.98  | 8.07  | 6.77  | 5.69 | 5.01  |
|                                   |                 | ROE (actual cap) | 29.8               | 32.2  | 58.0  | 99.8  | 93.2 | 72.4  |
|                                   |                 | ROA (adjusted)   | 6.19               | 3.36  | 6.51  | 6.85  | 5.51 | 4.38  |
|                                   |                 | ROE (adjusted)   | 103.2              | 56.0  | 108.5 | 114.2 | 91.8 | 73.0  |
| Avco National Bank                | 44 <sup>a</sup> | ROA (reported)   | N.A.               | 4.82  | 3.72  | 1.19  | N.A. | N.A.  |
|                                   |                 | ROE (actual cap) | N.A.               | 344.6 | 103.6 | 27.9  | N.A. | N.A.  |
|                                   |                 | ROA (adjusted)   | N.A.               | 4.23  | 4.66  | 5.06  | N.A. | N.A.  |
|                                   |                 | ROE (adjusted)   | N.A.               | 70.5  | 77.7  | 84.3  | N.A. | N.A.  |
| Call-report averages:             |                 | ROA (reported)   | 6.01               | 5.73  | 5.67  | 4.74  | 5.06 | 4.74  |
|                                   |                 | ROE (actual cap) | 56.8               | 115.7 | 85.2  | 81.6  | 78.9 | 55.6  |
|                                   |                 | ROA (adjusted)   | 4.44               | 4.55  | 5.66  | 5.77  | 5.13 | 4.47  |
|                                   |                 | ROE (adjusted)   | 74.0               | 75.9  | 94.3  | 96.2  | 85.5 | 74.6  |

Source: Consolidated reports of condition and income (call reports).

<sup>a</sup>Bank's credit card portfolio was acquired in 1986–1987; see Table 9.

**Figure 6:** Table 7: Call Report evidence on credit card profitability. Source: Ausubel (1991).

## 5.6 Table 9: portfolio resale premia

**Read:** Table 9 lists interbank sales of credit card accounts. The average reported premium is about 20%, with transactions ranging from 3% to 27%.



**Economic message.** Portfolio buyers are willing to pay more than outstanding balances because they expect excess profits from the acquired accounts. This directly contradicts the idea that profits are only accidental ex post realizations.

TABLE 9—PREMIA IN REALES OF CREDIT CARD ACCOUNTS

| Date           | Seller                          | Buyer             | Outstanding<br>balances | Premium<br>reported<br>(percentage) | Primary source        |
|----------------|---------------------------------|-------------------|-------------------------|-------------------------------------|-----------------------|
| April 1984     | Continental Bank                | Chemical Bank     | \$824 million           | 21                                  | WS April 2, 1984      |
| December 1986  | Texas American Bancshares       | Republic Bank     | \$50 million            | 14                                  | WS January 2, 1987    |
| December 1986  | Beneficial National Bank        | First Chicago     | \$1,100 million         | 13                                  | AB January 2, 1987    |
| February 1987  | National Bancshares Texas       | Lomas & Nettleton | \$41 million            | 14                                  | WS February 23, 1987  |
| April 1987     | Louisiana Bancshares            | Lomas & Nettleton | \$157 million           | 16                                  | WS April 16, 1987     |
| May 1987       | Avco National Bank              | Household Bank    | \$322 million           | 19                                  | WS April 29, 1987     |
| July 1987      | Bank of Mid-America             | Lomas & Nettleton | \$120 million           | 19                                  | WS July 23, 1987      |
| July 1987      | Colonial National Bank          | Household Bank    | \$317 million           | 11                                  | WS July 23, 1987      |
| September 1988 | First Republic Bank<br>Delaware | Citicorp          | \$623 million           | 25                                  | NY September 12, 1988 |
| November 1988  | Equibank                        | CoreStates Bank   | \$100 million           | 25                                  | KP                    |
| February 1989  | Meritor Financial Group         | Chase Manhattan   | \$85 million            | 24                                  | AB February 2, 1989   |
| March 1989     | Society for Savings<br>Bancorp  | First Chicago     | \$230 million           | 18                                  | NY April 13, 1989     |
| May 1989       | Michigan National Bank          | Chase Manhattan   | \$1,100 million         | 21                                  | WS July 19, 1989      |
| May 1989       | Empire of America               | Citicorp          | \$650 million           | 3                                   | AB June 2, 1989       |
| June 1989      | Colonial National Bank          | Household Bank    | \$98 million            | 25                                  | AB June 23, 1989      |
| July 1989      | Leader Federal Savings          | Chase Manhattan   | \$36 million            | 20                                  | KP                    |
| August 1989    | California Federal Bank         | Household Bank    | \$125 million           | 18                                  | AB September 18, 1989 |
| September 1989 | Chevy Chase Savings             | CoreStates Bank   | \$200 million           | 23                                  | AB September 18, 1989 |
| September 1989 | Imperial Savings & Loan         | Wells Fargo Bank  | \$280 million           | 22                                  | WS January 11, 1990   |
| September 1989 | Dreyfus Corp.                   | Bank of New York  | \$790 million           | 21                                  | AB September 20, 1989 |
| October 1989   | First City Bancorporation       | Bank of New York  | \$552 million           | 24                                  | AB October 19, 1989   |
| December 1989  | Bank South                      | Society National  | \$41 million            | 24                                  | AB December 27, 1989  |
| December 1989  | Bank of Boston                  | Chase Manhattan   | \$625 million           | 23                                  | AB January 5, 1990    |
| December 1989  | Investors Savings Bank          | Chase Manhattan   | \$24 million            | 25                                  | AB January 5, 1990    |
| January 1990   | Bank of New England             | Citicorp          | \$652 million           | 27                                  | NY January 30, 1990   |
| February 1990  | Colonial National Bank          | Household Bank    | \$50 million            | 20                                  | AB March 7, 1990      |
| April 1990     | Fleet/Norstar                   | Norwest           | \$200 million           | 20                                  | AB April 3, 1990      |

Sources: WS = *Wall Street Journal*; NY = *The New York Times*; AB = *The American Banker*; KP = Kidder Peabody (Anderson and Deans, 1989).

Figure 7: Table 9: Premia in resales of credit card accounts. Source: Ausubel (1991).

## 5.7 Table 10: capitalization of excess profits

**Read:** Table 10 asks what resale premium is justified under different assumptions about account lifetime, balance growth, and profitability. Large premia require returns several times ordinary banking returns unless one assumes very long lives and very fast growth.

**Economic message.** The portfolio-price evidence implies that market participants expected credit card profitability to persist. Premia are not easily reconciled with imminent convergence to competitive zero profits.

TABLE 10—PREMIA IN RESALE OF CREDIT CARD ACCOUNTS JUSTIFIED BY VARIOUS EXPECTED LIFETIMES OF ACCOUNTS, ANNUAL GROWTH RATES OF OUTSTANDINGS PER ACCOUNT, AND CREDIT CARD PROFITABILITY (AS A MULTIPLE OF ORDINARY RATE OF RETURN)

| Expected lifetime (years) | Multiple of ordinary rate of return | Premia (percentage) |                   |                   |                   |
|---------------------------|-------------------------------------|---------------------|-------------------|-------------------|-------------------|
|                           |                                     | 10-percent growth   | 20-percent growth | 30-percent growth | 40-percent growth |
| 2                         | 1                                   | 0.00                | 0.00              | 0.00              | 0.00              |
|                           | 2                                   | 2.40                | 2.51              | 2.62              | 2.73              |
|                           | 3                                   | 4.80                | 5.02              | 5.24              | 5.45              |
|                           | 4                                   | 7.20                | 7.53              | 7.85              | 8.18              |
|                           | 5                                   | 9.60                | 10.04             | 10.47             | 10.91             |
| 4                         | 1                                   | 0.00                | 0.00              | 0.00              | 0.00              |
|                           | 2                                   | 4.80                | 5.50              | 6.27              | 7.15              |
|                           | 3                                   | 9.60                | 10.99             | 12.55             | 14.29             |
|                           | 4                                   | 14.40               | 16.49             | 18.82             | 21.44             |
|                           | 5                                   | 19.20               | 21.98             | 25.10             | 28.58             |
| 6                         | 1                                   | 0.00                | 0.00              | 0.00              | 0.00              |
|                           | 2                                   | 7.20                | 9.05              | 11.38             | 14.30             |
|                           | 3                                   | 14.40               | 18.10             | 22.76             | 28.60             |
|                           | 4                                   | 21.60               | 27.15             | 34.15             | 42.90             |
|                           | 5                                   | 28.80               | 36.19             | 45.53             | 57.20             |

*Calculations:* For  $N$  = expected lifetime of credit card accounts,  $g$  = annual growth rate of outstandings per account (during lifetime of account),  $r = 0.10$  = interest rate used by bank in discounting,  $ROA_{cc}$  = return on assets from credit cards,  $ROA_{ord} = 0.012$  = ordinary return on assets in banking, and  $\Phi$  = premium in resale of credit card accounts (as proportion of outstanding balances at time of sale),

**Figure 8:** Table 10: Resale premia implied by persistent excess profitability. Source: Ausubel (1991).

## 5.8 Table 11: consumer behavior

**Read:** Table 11 reports that only about 21%–32% of customers avoid finance charges, meaning most active accounts incur interest. Average outstanding balances rise from about \$523 in 1979 to about \$1,038 in 1987.

**Economic message.** This table supports the behavioral mechanism. Consumers may behave as if they will not borrow, but most active customers actually do borrow and pay high finance charges.

TABLE 11—PERCENTAGE OF CUSTOMERS WHO AVOID FINANCE CHARGES  
AND AVERAGE OUTSTANDING BALANCES PER ACTIVE ACCOUNT

| Year | Percentage of customers<br>who avoid finance<br>charges | Number of banks<br>reporting percentage<br>of customers | Average outstanding<br>balance of<br>active account | Number of banks<br>reporting outstanding<br>balance |
|------|---------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|
| 1979 | 31.8                                                    | 6                                                       | \$523                                               | 7                                                   |
| 1980 | 28.7                                                    | 6                                                       | \$590                                               | 8                                                   |
| 1981 | 21.9                                                    | 6                                                       | \$660                                               | 8                                                   |
| 1982 | 21.0                                                    | 6                                                       | \$726                                               | 9                                                   |
| 1983 | 22.8                                                    | 8                                                       | \$711                                               | 10                                                  |
| 1984 | 21.9                                                    | 10                                                      | \$852                                               | 10                                                  |
| 1985 | 21.4                                                    | 12                                                      | \$1,014                                             | 12                                                  |
| 1986 | 24.6                                                    | 14                                                      | \$1,018                                             | 17                                                  |
| 1987 | 27.6                                                    | 9                                                       | \$1,038                                             | 10                                                  |

Source: Author's bank credit card survey (Appendix A, Table A1, question 8).

**Figure 9:** Table 11: Customers who avoid finance charges and average balances. Source: Ausubel (1991).

## 5.9 Table 12: implied competitive rates

**Read:** Table 12 computes the gap between actual credit card rates and implied competitive rates. The average gap for 1983–1988 is 4.88 percentage points; the 1987 gap is 5.02 percentage points.

**Economic message.** The economic magnitude is large. The paper's calculation suggests that competitive card rates would have been roughly five percentage points lower in the late 1980s, conditional on the paper's assumptions about profitability and balances accruing interest.

TABLE 12—IMPLIED DIFFERENTIAL BETWEEN “COMPETITIVE” CREDIT CARD  
INTEREST RATE AND ACTUAL RATE

| Year     | Number of banks<br>in sample | Adjusted return<br>on assets<br>(percentage) | Implied actual minus<br>“competitive” interest rate<br>(percentage) |
|----------|------------------------------|----------------------------------------------|---------------------------------------------------------------------|
| 1983     | 6                            | 4.44                                         | 3.60                                                                |
| 1984     | 13                           | 4.94                                         | 4.16                                                                |
| 1985     | 14                           | 6.23                                         | 5.59                                                                |
| 1986     | 15                           | 6.37                                         | 5.74                                                                |
| 1987     | 13                           | 5.72                                         | 5.02                                                                |
| 1988     | 10                           | 4.72                                         | 3.91                                                                |
| Average: | —                            | 5.59                                         | 4.88                                                                |

*Notes:* Adjusted return on assets is calculated by pooling the banks in Tables 6 and 7 for each year and calculating the arithmetic average of ROA (adjusted). The number of banks in the sample reflects one overlap in the years 1984–1988.

**Figure 10:** Table 12: Implied actual-minus-competitive credit card rates. Source: Ausubel (1991).

## 6 Short summary

- The paper studies a market that looks competitive by structure but not by outcomes.
- Credit card interest rates are almost flat around 18% despite large movements in funding costs.
- Regressions show tiny cost-of-funds pass-through: coefficients around 0.04–0.05.
- Major issuers earn adjusted pretax ROE far above ordinary banking returns during 1983–1988.
- Smaller-bank functional-cost evidence and portfolio-sale premia support the same profitability story.
- Resale premia around 20% indicate that buyers expect excess profits to persist.
- The paper’s mechanism emphasizes consumer inattention, search/switch frictions, and adverse selection.
- Consumers who do not expect to borrow may be the most profitable borrowers; intentional borrowers are more rate-sensitive and riskier.
- Lowering rates therefore attracts the wrong customers from the issuer’s perspective, weakening price competition.
- The policy implication is nuanced: laissez-faire arguments based on perfect competition are weak, but simple usury ceilings can also distort credit access.

## 7 Conclusion

Ausubel’s paper is a classic example of how a market with many firms can fail to deliver competitive outcomes. The evidence is not only that credit card interest rates are high. The deeper evidence is that rates are sticky relative to funding costs, profits are persistently far above ordinary banking returns, and market participants pay large premia for credit card portfolios because they expect those profits to continue.

The economic mechanism is demand-side and informational. Many consumers do not choose cards as if they expect to pay high finance charges, even though most active accounts do incur finance charges. This weakens the elasticity facing issuers. At the same time, the borrowers most likely to respond to lower rates are also the borrowers more likely to use credit lines heavily and default. Thus, rate cuts can worsen the applicant pool, making interest-rate competition unattractive.

The paper’s final lesson is methodological as much as substantive: market structure is only a starting point. To evaluate competition in consumer financial markets, one must look at price dynamics, profitability, customer switching, adverse selection, and the behavioral assumptions embedded in household choice.